T1

$\xi \sim R(0;\theta)$ $\quad \theta>0$

X_n - выборка

① $\tilde\theta_1 = 2\bar X = 2\cdot\frac{1}{n}\sum_{i=1}^n X_i$ ② $\tilde\theta_2 = X_{min}$ ③ $\tilde\theta_3 = X_{max}$ ④ $\tilde\theta_4 = X_1 + \frac{1}{n-1}\sum_{i=2}^n X_i$

$M\xi = \int_0^\theta x\cdot\frac{1}{\theta}\,dx = \frac{\theta}{2}$ $\quad M\xi^2 = \int_0^\theta x^2\cdot\frac{1}{\theta}\,dx = \frac{\theta^2}{3}$ $\quad D\xi = M\xi^2 - (M\xi)^2 = \frac{\theta^2}{12}$

① $\forall\theta>0$: $M\tilde\theta_1 = \frac{2}{n}\cdot\sum_{i=1}^n M\xi_i = \frac{2}{n}\cdot n\cdot M\xi = \theta \to$ несмещ.

$D\tilde\theta_1 = \frac{4}{n^2}\cdot n\cdot D\xi = \frac{4}{n}\cdot\frac{\theta^2}{12} \xrightarrow[n\to\infty]{} 0 \Rightarrow$ сост. (по дост. усл.)

② ~~$\forall\theta>0$~~

$\xi\sim F(x) \to \xi_{min} \sim \underbrace{1-(1-F(x))^n}_{\varphi(x)}$

$\varphi(x) = \varphi'(x) = n(1-F(x))^{n-1}F'(x) = n\left(1-\frac{x}{\theta}\right)^{n-1}\cdot\frac{1}{\theta}\ \{(0;\theta)\}$

$\forall\theta>0$: $MX_{min} = \int_0^\theta x\cdot n\left(1-\frac{x}{\theta}\right)^{n-1}\cdot\frac{1}{\theta}\,dx = \frac{\theta}{n+1} \to$ смещ.

$\tilde\theta_2' = (n+1)X_{min}$ - исправление

$M\tilde\theta_2' = (n+1)MX_{min} = \theta \to$ несмещ.

$D\tilde\theta_2' = (n+1)^2 DX_{min}$

$MX_{min}^2 = \int_0^\theta x^2 n\left(1-\frac{x}{\theta}\right)^{n-1}\cdot\frac{1}{\theta}\,dx = \frac{2\theta^2}{(n+1)(n+2)}$

$DX_{min} = \frac{2\theta^2}{(n+1)(n+2)} - \frac{\theta^2}{(n+1)^2} = \frac{\theta^2\cdot n}{(n+1)^2(n+2)} \to$ ничего не следует

$\forall\theta>0\ \forall\varepsilon>0\quad P(|\tilde\theta_2'-\theta|\geq\varepsilon) \xrightarrow[n\to\infty]{} 0$

$P(|\theta_2'-\theta|\geq\varepsilon) \geq P(\tilde\theta_2'\geq\theta+\varepsilon) = P((n+1)X_{min}\geq\theta+\varepsilon) = P_n(X_{min}\geq$
$\geq\frac{\theta+\varepsilon}{n+1}) = 1-\Phi\left(\frac{\theta+\varepsilon}{n+1}\right) = 1-\left(1-\left(1-\frac{\theta+\varepsilon}{(n+1)\cdot\theta}\right)^n\right) = \left(1-\frac{\theta+\varepsilon}{\theta(n+1)}\right)^n \to$
$\xrightarrow[n\to\infty]{} e^{-\frac{\theta+\varepsilon}{\theta}} > 0$

$P(|X_{min}-\theta|\geq\varepsilon)=P(X_{min}\leq\theta-\varepsilon)=\varphi(\theta-\varepsilon)=1-(1-F(\theta-\varepsilon))^n=$

$=1-(1-\frac{\theta-\varepsilon}{\theta})^n=1-(\frac{\varepsilon}{\theta})^n\xrightarrow[n\to\infty]{}1$

$\exists\varepsilon>0$ и $\exists\theta>0$: $\frac{\varepsilon}{\theta}<1$ → не зв. сост.

③ $X_{max}\sim\underbrace{(F(x))^n}_{\Psi(x)}$

$\varphi(x)=\Psi'(x)=n\left(\frac{x}{\theta}\right)^{n-1}\cdot\frac{1}{\theta}\{(0;\theta)\}$

$\mu X_{max}=\int_0^\theta x n\left(\frac{x}{\theta}\right)^{n-1}\cdot\frac{1}{\theta}dx=\frac{n}{n+1}\theta$ → смещ.

$\tilde\theta_3'=\frac{n+1}{n}X_{max}$ – исправление

$\mu\tilde\theta_3'=\frac{n+1}{n}\mu X_{max}=\theta$ → несмещ.

$\mu X_{max}^2=\int_0^\theta x^2 n\left(\frac{x}{\theta}\right)^{n-1}\cdot\frac{1}{\theta}dx=\frac{\theta^2 n}{n+2}$

$DX_{max}=\frac{\theta^2 n}{n+2}-\frac{\theta^2 n^2}{(n+1)^2}=\theta^2\cdot\frac{n}{(n+2)(n+1)^2}$

$D\tilde\theta_3'=\left(\frac{n+1}{n}\right)^2 DX_{max}\xrightarrow[n\to\infty]{}0$ → сост.

По опр. ($\tilde\theta_3$):

$P(|X_{max}-\theta|>\varepsilon)=P(\theta-X_{max}>\varepsilon)$

$P(X_{max}<\theta-\varepsilon)=\left(\frac{\theta-\varepsilon}{\theta}\right)^n\xrightarrow[n\to\infty]{}0$

$P(|X_{max}-\theta|>\varepsilon)\xrightarrow[n\to\infty]{}0$ → сост.

По опр. ($\tilde\theta_3'$):

$P\left(\left|\frac{n+1}{n}X_{max}-\theta\right|>\varepsilon\right)=P\left(\left|X_{max}+\frac{X_{max}}{n}-\theta\right|>\varepsilon\right)$

$\leq\frac{\theta}{n}\to0$

$\leq|X_{max}-\theta|+\frac{X_{max}}{n}\xrightarrow[n\to\infty]{}0$ → сост.

④ $\mu\tilde\theta_4=\mu X_1+\frac{1}{n-1}\sum_{i=2}^n\mu X_i=\mu g+\frac{1}{n-1}\sum_{i=2}^n\mu g=\frac{\theta}{2}+\frac{\theta}{2}=\theta$ → несмещ.

$D\tilde\theta_4=DX_1+\frac{1}{(n-1)^2}\sum_{i=2}^n DX_i=\frac{\theta^2}{12}+\frac{1}{n+1}\cdot\frac{\theta^2}{12}\not\xrightarrow[n\to\infty]{}0$ → ничего не следует.

g_n – незав., одинак. → ЗБЧ Хинчина

$X_1\xrightarrow{P}g$ $\frac{1}{n-1}\sum_{i=2}^n X_i\xrightarrow[n]{P}\mu X_1=\frac{\theta}{2}$

$\exists\mu g_n<\infty\to\frac{1}{n}\sum_{i=1}^n g_n\xrightarrow{P}\mu g$

$X_1+\frac{1}{n-1}\sum_{i=2}^n X_i\xrightarrow{P}g+\frac{\theta}{2}$ → не зв. сост.

Эффективность:

$D\tilde\theta_1=\frac{\theta^2}{3n}$ $\quad D\tilde\theta_3'=\frac{\theta^2}{(n+2)n}$

$\frac{1}{3n}$ $\quad\frac{1}{n(n+2)}$ $\quad n^2+2n>3n$, $n>1$ → $\tilde\theta_3'$ более эфф.